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Mounting mechanism for networking device

Abstract

A retention bracket for retaining an optical network component is provided that includes a base panel having a body defining cavity, and a mounting panel adapted to be at least partially inserted into the cavity of the base panel. The mounting panel includes a body having an upper portion, a lower portion, and a facing surface. The facing surface has at least one component coupling member extending therefrom that couples with the optical network component. The base panel and the mounting panel cooperate to receive and retain the at least one networking component.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to and the benefit of the filing date of U.S. Provisional Patent Application No. 63/714,422, filed on Oct. 31, 2024 and entitled “MOUNTING MECHANISM FOR NETWORKING DEVICE”, the entire disclosure of which is hereby expressly incorporated by reference herein.

FIELD OF THE DISCLOSURE

(1) This disclosure relates generally to networking systems, and, more particularly, to mounting mechanisms for networking systems.

BACKGROUND

(2) A conventional passive optical network (PON) includes one or more optical line terminals (OLTs) at a central location connecting to one or more optical last mile termination units (LMTUs) disposed at respective customer premises (e.g., physical locations serviced by the PON) via one or more optical fibers. A PON is typically implemented using a point-to-multipoint topology in which a feeder optical fiber from an OLT serves multiple last mile termination units. In some examples, an LMTU may be in the form of an optical network terminal (ONT) or an optical network unit (ONU) that is optically connected to the OLT via a respective distribution optical fiber received at the LMTU. Such ONTs are typically disposed within enclosures that may be affixed to a structure, a pole, and/or other suitable stationary objects.

(3) Because of continued advancements to PON technologies, improved and/or higher-powered ONTs have been developed which may be of different dimensions than existing units. Such ONTs may require replacement retention mechanisms (e.g., brackets, straps, etc.) in order to securely retain the ONT within the enclosure. Such mechanisms may require on-site modifications and/or alterations to the existing enclosure (e.g., by removing portions of the enclosure or other retention features), and ultimately may not provide for securely retaining the ONT. Further, such modifications may be time-consuming and may lead to improper installations. Further still, such modifications to the existing enclosure may result in the enclosure no longer being capable of securely retaining other ONT models due to the removal of necessary retention features and/or material.

(4) Accordingly, there is a need for improved devices having improved functionalities.

SUMMARY

(5) In accordance with a first aspect, a retention bracket for retaining an optical network component

is provided that includes a base panel having a body defining cavity, and a mounting panel adapted to be at least partially inserted into the cavity of the base panel. The mounting panel includes a body having an upper portion, a lower portion, and a facing surface. The facing surface has at least one component coupling member extending therefrom that couples with the optical network component. The base panel and the mounting panel cooperate to receive and retain the at least one networking component.

(6) In some examples, the at least one component coupling member includes a keyhole protrusion adapted to engage a corresponding keyhole opening formed on the optical network component. Further, in some approaches, the mounting panel is adapted to be slidably inserted into the cavity from a cavity opening formed on a side portion of the base panel.

(7) In some arrangements, at least one of the upper portion or the lower portion of the mounting panel includes a sawtooth profile adapted to be received by a corresponding sawtooth opening formed in the cavity of the base panel. Further, the sawtooth profile may be in the form of a crenellated sawtooth profile.

(8) In some implementations, at least one of the upper portion or the lower portion of the mounting panel further includes a chamfered surface adapted to slidably engage with the body of the base panel. The body of the base panel may further include a chamfered surface adapted to slidably engage the chamfered surface of the mounting panel.

(9) In these and other forms, the base panel may include a stop member that engages the mounting panel to limit lateral movement thereof. Further, in some arrangements, the body of the mounting panel includes at least one through hole that receives a fastener to couple the mounting panel with a structure.

(10) In accordance with a second aspect, an enclosure for receiving at least one networking component may include a shell defining an interior volume and a retention bracket adapted to be received within the interior volume of the shell. The retention bracket includes a base panel having a body defining a cavity and a mounting panel adapted to be at least partially inserted into the cavity of the base panel. Further, the mounting panel includes a body having an upper portion, a lower portion, and a facing surface having at least one component coupling member extending therefrom adapted to couple with the optical network component. The base panel, and the mounting panel cooperate to receive and retain the at least one networking component within the shell.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views, together with the detailed description below, are incorporated in and form part of the specification, and serve to further illustrate examples of concepts that include the claimed invention, and explain various principles and advantages of those examples.

(2) FIG. 1 is a perspective view of an example enclosure for retaining at least one networking component in which the systems, methods, and/or techniques of the present disclosure may be implemented.

(3) FIG. 2 is a perspective view of a portion of the example enclosure of FIG. 1 in a partially-assembled configuration in accordance with various examples.

(4) FIG. 3 is a perspective view of a portion of the example enclosure of FIGS. 1 & 2 in accordance with various examples.

(5) FIG. 4 is a perspective view of an example retention bracket having an example base panel and a first example mounting panel for use with the example enclosure of FIGS. 1-3 in accordance with various examples.

(6) FIG. 5 is a front elevation view of the example retention racket of FIG. 4 in accordance with various examples.

(7) FIG. 6 is a right side elevation view of the example retention racket of FIGS. 4 & 5 in accordance with various examples.

(8) FIG. 7 is a perspective view of the example base panel of the example retention bracket of FIGS. 4-6 in accordance with various examples.

(9) FIG. 8 is a right side perspective view of the example base panel of FIGS. 4-7 in accordance with various examples.

(10) FIG. 9 is a right side elevation view of the example base panel of FIGS. 4-8 in accordance with various examples.

(11) FIG. 10 is a perspective view of the first example mounting panel of the example retention bracket of FIGS. 4-6 in accordance with various examples.

(12) FIG. 11 is a rear perspective view of the first example mounting panel of the example retention bracket of FIGS. 4-6 in accordance with various examples.

(13) FIG. 12 is a perspective view of a second example mounting panel for use with the example retention bracket of FIGS. 4-6 in accordance with various examples.

(14) FIG. 13 is a perspective view of an example enclosure having an example base panel coupled therewith in an alternative location in accordance with various examples.

(15) FIG. 14 is a perspective view of the example enclosure of FIG. 13 having the second example mounting panel of FIG. 12 coupled therewith in accordance with various examples.

(16) FIG. 15 is a perspective view of a third example mounting panel for use with the example retention bracket of FIGS. 4-6 in accordance with various examples.

(17) Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions and/or relative positioning of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of various examples. Also, common but well-understood elements that are useful or necessary in a commercially feasible examples are often not depicted in order to facilitate a less obstructed view of these various examples. It will further be appreciated that certain actions and/or steps may be described or depicted in a particular order of occurrence while those skilled in the art will understand that such specificity with respect to sequence is not actually required. It will also be understood that the terms and expressions used herein have the ordinary technical meaning as is accorded to such terms and expressions by persons skilled in the technical field as set forth above except where different specific meanings have otherwise been set forth herein.

(18) The apparatus and method components have been represented where appropriate by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding examples of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

(19) Although the figures show parts with clean lines and boundaries, some or all of these lines and/or boundaries may be idealized. In reality, the boundaries and/or lines may be unobservable, blended, and/or irregular. Use of terms such as up, down, top, bottom, side, end, front, back, etc. herein are used with reference to a currently considered or illustrated orientation. If they are considered with respect to another orientation, it should be understood that such terms must be correspondingly modified.

DETAILED DESCRIPTION

(20) Disclosed examples of the disclosure provide a number of advantages over existing techniques for securing and retaining components of passive optical networks (PONs). Generally speaking, a component of a PON, as utilized herein, may be in the form of a last mile termination unit (e.g., an optical network terminal (ONT) or an optical network unit (ONU) disposed at customer premises).

Such a component may be securely retained within an existing enclosure using a bracket or locking apparatus. More specifically, the brackets described herein may accommodate dimensionally larger networking components without the need for modifications to the enclosure and/or the networking component to achieve a secure fit. Further, in some examples, the brackets described herein may eliminate the need for additional securing mechanisms. Last, in some examples, the brackets described herein may be adjustable in nature to securely accommodate and retain various networking components having different sizes and/or shapes.

(21) The retention mechanisms described herein incorporate a universal base panel that is coupled with an enclosure or other fixture. This base panel may receive and/or otherwise couple with any number of unique mounting panels having specific orientations and features to assist with coupling to desired networking components. The apparatuses described herein may advantageously be installed within an enclosure having limited space due to the novel arrangement of the engagement mechanisms between the base panel and the mounting panel.

(22) Turning now to the Figures, an enclosure **100** is provided for retaining at least one networking component **10** (e.g., an ONT, ONU, and/or any other device). The enclosure **100** may include a shell **102** defining an interior volume **102a** and a retention bracket **120**. More specifically, the illustrated shell **102** of the enclosure **100** includes a rear panel **103a**, a number of sidewalls **103b**, and a door **103c**. While the illustrated enclosure **100** is in the form of a generally rectangular box, it is to be appreciated that the enclosure **100** may take any number of suitable shapes, forms, and/or configurations.

(23) The retention bracket **120** may be at least partially disposed within, coupled with, and/or otherwise positionable within the interior volume **102a** of the shell **102**. The retention bracket **120** includes a base panel **122** and a mounting panel **142** adapted to operably (e.g., slidably) couple with each other. The various components of the retention bracket **120** may be constructed from any number of suitable materials such as, for example, polymeric materials, metals, and the like. Other examples are possible.

(24) Generally speaking, the base panel **122** is adapted to be removably or permanently affixed to or otherwise formed with the shell **102**. With reference to FIGS. 3-9, the base panel **122** includes a body **124** defining a cavity **124a**. Positioned on a side portion **124b** of the cavity **124a** is a cavity opening **124c**. In other words, in the illustrated examples, the cavity **124a** is in the form of a depression or recess formed in the body **124** that extends to an edge thereof.

(25) In some examples, the base panel **122** may be secured with the shell via fasteners, latching engagements, and the like. As a non-limiting example, as can be seen in FIG. 3, the base panel **122** may include any number of through holes **126** that may receive a fastener to secure with the shell **102**. It is to be appreciated that in some forms, the base panel **122** may be coupled with the shell **102** via adhesives, fusion welding, and the like, and in yet other examples, the base panel **122** may be integrally formed with the shell **102**. In other examples (not illustrated) a portion of the shell **102** may include a recessed portion that is dimensioned to receive at least a portion of the base panel **122** therein. In some examples, the recessed portion may frictionally engage the base panel **122**, while in other examples, the coupling may be aided by the use of fasteners or other suitable coupling approaches.

(26) With particular reference to FIGS. 7-9, the body **124** of the retention panel **122** includes a sawtooth profile **128** whereby a number of abutments **129** extend inwardly into the cavity **124a**. Each adjacent abutment **129** is separated by a corresponding gap **129a**. In the illustrated examples, the abutments **129** are in the form of a crenellated sawtooth profile whereby a number of rectangular protrusions and gaps are formed. It is to be appreciated that any number of suitable shapes or configurations for the sawtooth profile **128** may be used whereby any number of desired abutments **129** and corresponding openings or gaps **129a** are formed.

(27) In some arrangements, any number of the abutments **129** may include an inner chamfered surface **130**. This chamfered surface **130** may be angled relative to the outer surface of the

abutment **129**. In some arrangements, an angle of approximately 45° may be formed between the angled surface **130** and a rear wall **132** of the base panel **122**.

(28) The base panel **122** further includes a stop member **136**. In some examples, the stop member **136** may be formed on the rear wall **132** of the base panel **122** and may be in the form of a bump, nodule, or similar protrusion. Other examples are possible.

(29) With reference to FIGS. **4-6 & 10**, the first example mounting panel **142** is provided. The mounting panel **142** includes a body **144** having an upper portion **144a**, a lower portion **144b**, a device facing surface **144c**, and a base panel facing surface **144d**. The device facing surface **144c** includes any number of component coupling members **152**. In the illustrated example, each component coupling member **152** is in the form of a keyhole protrusion **154** that extends from a finger **156** projecting from the device facing surface **144c**. The keyhole protrusion **154** may be of any suitable size and/or configuration to operably couple with a corresponding keyhole opening **12** disposed on the networking component **10** (see, e.g., FIG. **3**). The keyhole protrusion **154** may be inserted into the corresponding keyhole opening **12** of the networking component **10** to secure the networking component **10** with the mounting panel **142**. As a result, the networking component **10** may rest against the device facing surface **144c** while being additionally supported by the component coupling member **152**.

(30) Like the body **124** of the base panel **122**, the body **144** of the mounting panel **142** may also include a sawtooth opening or profile **148**. More specifically, the upper portion **144a** and/or the lower portion **144b** of the body **144** may include a number of abutments **149** that protrude outwardly from the respective upper and lower portions **144a**, **144b**. Each adjacent abutment **149** is separated by a corresponding gap **149a**. In the illustrated examples, the abutments **149** are in the form of a crenellated sawtooth profile whereby a number of rectangular protrusions and gaps are formed. It is to be appreciated that any number of suitable shapes or configurations for the sawtooth profile **148** may be used whereby any number of desired abutments **149** and corresponding openings or gaps **149a** are formed.

(31) In some arrangements, any number of the abutments **149** may include a chamfered surface **150**. This chamfered surface **150** may be angled relative to the device facing surface **144c**. In some arrangements, an angle of approximately 45° may be formed between the chamfered surface **150** and the device facing surface **144c**. Notably, and as will be discussed in further detail below, the chamfered surfaces **150** formed on the abutments **149** of the mounting panel **142** are arranged to abut the corresponding chamfered surface **130** formed on the abutments **129** of the base panel **122**.

(32) The mounting panel **142** may further include a stop member **146** in the form of an indentation or protrusion. In some examples, the stop member **146** may be formed on the base panel facing surface **144d** such that it may engage with the corresponding stop member **136** formed on the base panel **122**. Other examples are possible.

(33) The mounting panel **142** may be coupled with the base panel **122** in a number of approaches. Generally, the mounting panel **142** is dimensioned to be slidably inserted into the cavity **124a** of the base panel **122**. The mounting panel **142** may be slidably inserted into the cavity **124a** via the cavity opening **124c** positioned at the side portion **124b** thereof, and may be inserted into the cavity **124a** until the stop member **136** engages the stop member **146** of the mounting panel **142** (and/or when the mounting panel **142** is fully inserted into the cavity **124a** and contacts a sidewall of the body **124**). In another approach and in environments where the mounting panel **142** may not be fully insertable into the cavity **124a** via the cavity opening **124c** (e.g., in environments where the enclosure **100** has components and/or dimensions that block or prevent such installation), the mounting panel **142** may be coupled with the base panel **122** by positioning each abutment **149** of the body **144** of the mounting panel **142** within a corresponding gap **129a** of the abutment **129** of the body **124** of the base panel **122**. In this initial arrangement, the openings **149a** of the abutment **149** of the body **144** of the mounting panel **142** receive the corresponding abutment **129** of the body **124** of the base panel **122** (see, e.g., FIG. **3**). The mounting panel **142** may be fully inserted

into the cavity **124a** of the body **124** such that the base panel facing surface **144d** contacts or is otherwise positioned adjacent to the rear wall **132**, and the mounting panel **142** may then be moved to the fully-coupled position. Such an arrangement requires significantly less lateral clearance (e.g., between approximately 2 cm and approximately 10 cm) than inserting the mounting panel **142** through the cavity opening **124c**.

(34) In still other implementations, the cavity **124a** may be dimensioned with an appropriate tolerance to allow the mounting panel **142** to be rotatably inserted into the cavity **124a** of the base panel **122**. More specifically, the corresponding upper (or lower) chamfered surfaces **130**, **150** may first be positioned adjacent to each other, whereupon the bottom (or upper) portion of the mounting panel **142** may be rotated into the coupled position where both the upper and lower chamfered surfaces **130**, **150** abut each other. Such an arrangement requires minimal lateral clearance, as the mounting panel **142** is merely rotated or otherwise swung into place without sliding the mounting panel **142** within the cavity **124a**.

(35) In any of these approaches, the rear wall **132** of the base panel **122** abuts the base panel facing surface **144d** of the mounting panel **142**. Additionally, the chamfered surface **130** of each abutment **129** of the base panel **122** contacts the chamfered surface **150** of each abutment **149** of the mounting panel **142**. Such engagement between the abutments **129**, **149** causes the mounting panel **142** to be retained by the base panel **122**, and as a result, also causes the networking component **10** to be retained by the base panel **122**.

(36) It is to be appreciated that in some arrangements, the networking component **10** may be coupled with the mounting panel **142** before the mounting panel **142** is slidably coupled with the base panel **122**. In other examples, the networking component **10** may be coupled with the mounting panel **142** after the mounting panel **142** is slidably coupled with the base panel **122**.

(37) As previously noted, the retention bracket **120** described herein may incorporate a universal base panel **122** that may receive a number of different mounting panels. In some arrangements (not illustrated), mounting panels may be used having component coupling members positioned at alternative locations to accommodate differently-dimensioned networking components. Further, with reference to FIGS. **12-15**, different mounting panels may be provided having additional features specific to the components being retained. For example, FIGS. **12-14** depict an alternative enclosure **200** having a base panel **122** coupled to a sidewall **203b** thereof. It is to be appreciated that the features of the enclosure **200** are similar to those of the enclosure **100**, and as such, for the sake of brevity, these features will not be discussed in additional detail. Such similar features will have similar reference numerals as those described in the above example, but are provided in multiples of 100 (e.g., 200, 300). In a first example, the mounting panel **242** includes a component coupling member **252** in the form of an elongated arm that is dimensioned to frictionally engage a networking component (e.g., a power supply or similar component). Similarly, FIG. **15** illustrates a third example mounting panel **342** having a component coupling member **352** in the form of two arms that may frictionally engage a networking component.

(38) So arranged, the enclosure **100** and the retention bracket **120** thereof may be used to safely retain larger networking components. As such, the retention bracket **120** may be used with existing shells **102** without needing to make modifications to the shell **102** or the networking device **10**. Put differently, the retention bracket **120** may be provided as a retrofittable component.

(39) In the foregoing specification, specific embodiments have been described. However, one of ordinary skill in the art appreciates that various modifications and changes can be made without departing from the scope of the invention as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of present teachings. Numerous alternative examples could be implemented, using either current technology or technology developed after the filing date of this patent, which would still fall within the scope of the claims. Additionally, the described embodiments/examples/implementations should not be interpreted as

mutually exclusive and should instead be understood as potentially combinable if such combinations are permissive in any way. In other words, any feature disclosed in any of the aforementioned embodiments/examples/implementations may be included in any of the other aforementioned embodiments/examples/implementations.

(40) The benefits, advantages, solutions to problems, and any element(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential features or elements of any or all the claims. The claimed invention is defined solely by the appended claims including any amendments made during the pendency of this application and all equivalents of those claims as issued.

(41) Moreover, in this document, relational terms such as first and second, top and bottom, and the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” “has,” “having,” “includes,” “including,” “contains,” “containing” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises, has, includes, contains a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a”, “has . . . a”, “includes . . . a”, “contains . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises, has, includes, contains the element. The terms “a” and “an” are defined as one or more unless explicitly stated otherwise herein. The terms “substantially”, “essentially”, “approximately”, “about” or any other version thereof, are defined as being close to as understood by one of ordinary skill in the art, and in one non-limiting example the term is defined to be within 10%, in another example within 5%, in another example within 1% and in another example within 0.5%. The term “coupled” as used herein is defined as connected, although not necessarily directly and not necessarily mechanically. A device or structure that is “configured” in a certain way is configured in at least that way, but may also be configured in ways that are not listed.

(42) Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, “A, B or C” refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, and (7) A with B and with C. As used herein, the phrase “at least one of A and B” is intended to refer to any combination or subset of A and B such as (1) at least one A, (2) at least one B, and (3) at least one A and at least one B. Similarly, the phrase “at least one of A or B” is intended to refer to any combination or subset of A and B such as (1) at least one A, (2) at least one B, and (3) at least one A and at least one B.

(43) The Abstract of the Disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various examples for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed examples require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter may lie in less than all features of a single disclosed example. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

(44) By way of example, and not limitation, the disclosure herein contemplates at least the following examples: 1. A retention bracket adapted to retain an optical network component, the retention bracket including a base panel having a body defining cavity, and a mounting panel adapted to be at least partially inserted into the cavity of the base panel. The mounting panel includes a body having an upper portion, a lower portion, and a facing surface. The facing surface

has at least one component coupling member extending therefrom adapted to couple with the optical network component. The base panel and the mounting panel cooperate to receive and retain the at least one networking component. 2. The retention bracket of example 1, wherein the at least one component coupling member includes a keyhole protrusion adapted to engage a corresponding keyhole opening formed on the optical network component. 3. The retention bracket of example 1 or 2, wherein the mounting panel is adapted to be slidably inserted into the cavity from a cavity opening formed on a side portion of the base panel. 4. The retention bracket of any one of examples 1-3, wherein at least one of the upper portion or the lower portion of the mounting panel includes a sawtooth profile adapted to be received by a corresponding sawtooth opening formed in the cavity of the base panel. 5. The retention bracket of example 4, wherein the sawtooth profile comprises a crenellated sawtooth profile. 6. The retention bracket of any one of examples 1-5, wherein at least one of the upper portion or the lower portion of the mounting panel further includes a chamfered surface adapted to slidably engage with the body of the base panel. 7. The retention bracket of example 6, wherein the body of the base panel further includes a chamfered surface adapted to slidably engage the chamfered surface of the mounting panel. 8. The retention bracket of any one of examples 1-7, wherein the base panel includes a stop member adapted to engage the mounting panel to limit lateral movement thereof. 9. The retention bracket of any one of examples 1-8, wherein the body of the mounting panel includes at least one through hole adapted to receive a fastener to couple the mounting panel with a structure. 10. An enclosure adapted to receive at least one optical networking component, the enclosure comprising a shell defining an interior volume and a retention bracket adapted to be received within the interior volume of the shell. The retention bracket includes a base panel having a body defining cavity, and a mounting panel adapted to be at least partially inserted into the cavity of the base panel. The mounting panel includes a body having an upper portion, a lower portion, and a facing surface having at least one component coupling member extending therefrom adapted to couple with the optical network component. wherein the base panel, and the mounting panel cooperate to receive and retain the at least one networking component within the shell. 11. The enclosure of example 10, wherein the shell includes a cavity adapted to at least partially receive a portion of the body of the base panel. 12. The enclosure of example 10 or 11, wherein the base panel is integrally formed with the shell. 13. The enclosure of any one of examples 10-12, wherein the at least one component coupling member includes a keyhole protrusion adapted to engage a corresponding keyhole opening formed on the optical network component. 14. The enclosure of any one of examples 10-13, wherein the mounting panel is adapted to be slidably inserted into the cavity from a cavity opening formed on a side portion of the base panel. 15. The enclosure of any one of examples 10-14, wherein at least one of the upper portion or the lower portion of the mounting panel includes a sawtooth profile adapted to be received by a corresponding sawtooth opening formed in the cavity of the base panel. 16. The enclosure of example 15, wherein the sawtooth profile comprises a crenellated sawtooth profile. 17. The enclosure of any one of examples 10-16, wherein at least one of the upper portion or the lower portion of the mounting panel further includes a chamfered surface adapted to slidably engage with the body of the base panel. 18. The enclosure of example 17, wherein the body of the base panel further includes a chamfered surface adapted to slidably engage the chamfered surface of the mounting panel. 19. The enclosure of any one of examples 10-18, wherein the base panel includes a stop member adapted to engage the mounting panel to limit lateral movement thereof. 20. The of any one of examples 10-19, wherein the body of the mounting panel includes at least one through hole adapted to receive a fastener to couple the mounting panel with a structure

(45) Additionally, the benefits, advantages, solutions to problems, and any element(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential features or elements of any or all the claims.

(46) Finally, any references, including, but not limited to, publications, patent applications, and patents cited herein are hereby incorporated in their entirety by reference to the same extent as if

each reference were individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein.

(47) The patent claims at the end of this patent application are not intended to be construed under 35 U.S.C. § 112(f) unless traditional means-plus-function language is expressly recited, such as “means for” or “step for” language being explicitly recited in the claim(s). The systems and methods described herein are directed to an improvement to computer functionality, and improve the functioning of conventional computers.

(48) Although certain example methods, apparatus and articles of manufacture have been disclosed herein, the scope of coverage of this patent is not limited thereto. On the contrary, this patent covers all methods, apparatus and articles of manufacture fairly falling within the scope of the claims of this patent.

Claims

1. A retention bracket adapted to retain an optical network component, the retention bracket including: a base panel having a body defining a cavity, the body including at least one abutment extending inwardly into the cavity and defining a sawtooth profile, the at least one abutment having an inner chamfered surface; and a mounting panel adapted to be at least partially inserted into the cavity of the base panel, the mounting panel including a body having an upper portion, a lower portion, and a facing surface, the facing surface having at least one component coupling member extending therefrom adapted to couple with the optical network component, at least one of the upper portion or the lower portion including at least one abutment defining a sawtooth profile, the at least one abutment having a chamfered surface angled relative to the facing surface; wherein the base panel and the mounting panel cooperate to receive and retain the optical network component via sliding engagement between the at least one abutment of the base panel and the at least one abutment of the mounting panel.
2. The retention bracket of claim 1, wherein the at least one component coupling member includes a keyhole protrusion adapted to engage a corresponding keyhole opening formed on the optical network component.
3. The retention bracket of claim 1, wherein the mounting panel is adapted to be slidably inserted into the cavity from a cavity opening formed on a side portion of the base panel.
4. The retention bracket of claim 1, wherein the sawtooth profile comprises a crenellated sawtooth profile.
5. The retention bracket of claim 1, wherein the base panel includes a stop member adapted to engage the mounting panel to limit lateral movement thereof.
6. The retention bracket of claim 1, wherein the body of the mounting panel includes at least one through hole adapted to receive a fastener to couple the mounting panel with a structure.
7. An enclosure adapted to receive at least one optical network component, the enclosure comprising: a shell defining an interior volume; and a retention bracket adapted to be received within the interior volume of the shell, the retention bracket including: a base panel having a body defining a cavity, the body including at least one abutment extending inwardly into the cavity and defining a sawtooth profile, the at least one abutment having an inner chamfered surface; and a mounting panel adapted to be at least partially inserted into the cavity of the base panel, the mounting panel including a body having an upper portion, a lower portion, and a facing surface, the facing surface having at least one component coupling member extending therefrom adapted to couple with the optical network component, at least one of the upper portion or the lower portion including at least one abutment defining a sawtooth profile, the at least one abutment having a chamfered surface angled relative to the facing surface; wherein the base panel, and the mounting panel cooperate to receive and retain the optical network component within the shell via sliding engagement between the at least one abutment of the base panel and the at least one abutment of

the mounting panel.

8. The enclosure of claim 7, wherein the shell includes a cavity adapted to at least partially receive a portion of the body of the base panel.

9. The enclosure of claim 7, wherein the base panel is integrally formed with the shell.

10. The enclosure of claim 7, wherein the at least one component coupling member includes a keyhole protrusion adapted to engage a corresponding keyhole opening formed on the optical network component.

11. The enclosure of claim 7, wherein the mounting panel is adapted to be slidably inserted into the cavity from a cavity opening formed on a side portion of the base panel.

12. The enclosure of claim 7, wherein the sawtooth profile comprises a crenellated sawtooth profile.

13. The enclosure of claim 7, wherein the base panel includes a stop member adapted to engage the mounting panel to limit lateral movement thereof.

14. The enclosure of claim 7, wherein the body of the mounting panel includes at least one through hole adapted to receive a fastener to couple the mounting panel with a structure.
